

KOKEMAKI TULKKILA, Finland

WMO: 029370

Lat: 61.2531N

Lon: 22.3461E

Elev: 39

StdP: 100.86

Time Zone: 2.00 (EUE)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -20.9	(c) -17.8	(d) -23.2	(e) 0.5	(f) -20.8	(g) -20.1	(h) 0.6	(i) -17.7	(j) 9.5	(k) -0.8	(l) 8.6	(m) -1.2	(n) 1.2	(o) 130	(p) 0.620

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 10.1	(c) 27.1	(d) 18.1	(e) 24.9	(f) 17.3	(g) 23.0	(h) 16.4	(i) 19.6	(j) 24.6	(k) 18.5	(l) 23.3	(m) 17.5	(n) 21.8	(o) 3.5	(p) 190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) 18.0	(b) 13.0	(c) 21.1	(d) 16.6	(e) 11.8	(f) 20.2	(g) 15.6	(h) 11.1	(i) 19.3	(j) 55.9	(k) 24.7	(l) 52.4	(m) 23.5	(n) 49.3	(o) 22.0	(p) 22.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
8.2	7.2	6.4	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
			-21.7	28.8	4.7	2.9	-25.1	30.9	-27.8	32.6	-30.5	34.3	-33.9	36.4
			-21.9	20.4	4.6	1.5	-25.2	21.5	-28.0	22.4	-30.6	23.3	-33.9	24.4
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	5.6	-6.0	-3.5	-1.5	4.3	10.6	13.9	16.7	15.5	11.1	4.9	2.0	-1.8
	DBStd	8.64	6.41	5.20	4.76	3.29	4.37	3.35	3.03	2.76	3.39	3.73	3.42	5.00
	HDD10.0	2251	497	377	357	174	45	5	0	0	27	162	241	365
	HDD18.3	4694	755	610	616	421	242	138	69	95	217	417	491	623
	CDD10.0	632	0	0	0	3	63	122	208	171	61	4	0	0
	CDD18.3	34	0	0	0	0	2	5	19	8	1	0	0	0
	CDH23.3	378	0	0	0	0	44	59	205	68	1	0	0	0
	CDH26.7	74	0	0	0	0	4	8	48	14	0	0	0	0
(14) Wind	WSAvg	3.1	3.0	3.2	3.3	3.0	3.2	3.3	2.9	3.0	2.9	3.1	3.2	3.3
(15) Precipitation	PrecAvg	628	49	34	33	35	38	62	70	69	59	65	60	56
	PrecMax	828	93	75	62	63	98	121	124	211	126	146	129	108
	PrecMin	506	14	2	8	5	8	16	4	20	14	8	11	19
	PrecStd	70	19	18	13	15	22	23	32	39	34	35	24	22
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.7	5.7	10.7	20.3	27.1	28.0	30.4	29.0	22.5	14.5	10.6	9.3
		MCWB	3.6	3.2	5.7	11.2	15.9	19.0	19.8	20.5	16.4	11.7	9.7	8.4
	2%	DB	3.5	4.3	8.0	16.2	24.2	24.9	27.8	25.0	20.2	12.5	8.7	6.2
		MCWB	2.7	2.9	4.5	9.4	14.7	17.2	18.9	18.0	15.5	10.9	8.0	5.0
	5%	DB	2.2	3.2	5.6	13.2	21.9	22.7	25.6	22.9	18.3	11.7	7.5	4.7
		MCWB	1.6	2.3	3.1	7.5	13.7	16.0	18.2	16.8	14.3	10.4	6.9	3.9
	10%	DB	1.3	2.3	4.0	10.7	19.4	21.0	23.2	21.2	16.8	10.6	6.5	3.6
		MCWB	0.8	1.6	2.1	6.3	12.4	14.6	17.1	16.0	13.8	9.4	5.9	2.9
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.0	3.8	6.6	12.3	17.6	20.0	21.7	20.9	17.4	12.4	10.0	8.5
		MCDB	4.5	4.8	9.4	17.6	23.7	25.0	27.5	25.8	21.1	13.8	10.4	9.3
	2%	WB	2.9	3.1	4.7	10.0	16.0	18.2	20.1	18.8	16.0	11.5	8.2	5.4
		MCDB	3.4	3.9	7.4	16.2	22.1	23.4	25.1	23.0	19.0	12.4	8.6	6.2
	5%	WB	1.8	2.5	3.4	8.0	14.6	16.9	19.0	17.8	15.1	10.7	7.1	4.0
		MCDB	2.2	3.3	5.2	12.3	20.6	21.7	24.2	21.7	17.5	11.5	7.5	4.7
	10%	WB	0.9	1.7	2.4	6.7	13.0	15.7	18.0	16.8	14.2	9.6	6.1	3.1
		MCDB	1.3	2.4	3.8	10.2	19.0	19.8	22.6	20.0	16.2	10.5	6.5	3.6
(35) Mean Daily Temperature Range	5% DB	MDBR	5.7	5.3	7.6	9.5	11.2	9.5	10.1	9.6	8.5	6.2	3.7	4.4
		MCDBR	3.5	5.2	8.8	14.9	15.8	13.2	14.0	12.9	11.1	6.7	4.5	4.9
		MCWBR	3.3	4.3	6.0	8.8	7.6	6.7	6.3	6.5	6.8	5.4	4.3	4.4
	5% WB	MCDDBR	3.5	4.7	7.7	13.1	13.2	11.5	11.9	10.5	9.3	5.6	4.6	4.6
		MCWBR	3.4	4.0	5.5	8.1	6.8	6.4	5.9	5.8	6.1	5.1	4.3	4.3
(40) Clear-Sky Solar Irradiance	taub		0.263	0.306	0.329	0.368	0.364	0.356	0.384	0.370	0.344	0.327	0.285	0.359
	taud		2.130	2.224	2.308	2.330	2.405	2.451	2.396	2.426	2.489	2.398	2.199	2.564
	Ebn at Noon		452	662	787	823	859	872	833	815	774	644	420	300
	Edh at Noon		38	67	86	103	103	100	104	93	73	57	36	24
(44) All-Sky Solar Radiation	RadAvg		0.22	0.78	2.27	3.75	5.14	5.49	5.14	4.03	2.42	1.03	0.27	0.12
	RadStd		0.04	0.17	0.28	0.41	0.52	0.41	0.51	0.42	0.30	0.19	0.05	0.01

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (57 neighbors)		+0.45	N/A	N/A	N/A	N/A	N/A	-135	N/A	N/A	N/A

Nomenclature: See separate page